Measurement & Angles: Modulo-60 Clock Navigation

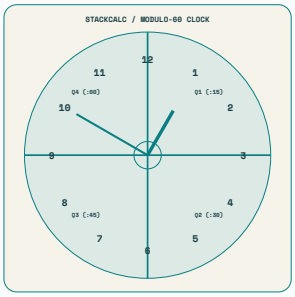
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.4.MD.C.5

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Clock Dial

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Rotate clock minute hand by 15-minute quadrant zones. One full revolution is 360° / 60 min.

Sketch Model on Paper:
Draw clock dial partitioned into four 90° quadrants (15, 30, 45, 60 min).

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	25	X: 25	Minutes elapsed (25)
2.	ENTER	Y: 25 | X: 25	Stack lift
3.	6	Y: 25 | X: 6	6° per minute into X
4.	×	X: 150	Multiply: 25 × 6° =

EXPECTED LCD X: 150°

1 How many degrees does minute hand rotate in 10 minutes?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 How many degrees does minute hand rotate in 20 minutes?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 How many degrees does minute hand rotate in 50 minutes?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 A minute hand turns 180°. How many minutes passed?

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 4

Measurement & Angles: Modulo-60 Clock Navigation • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: 4-Quadrant Modulo-60 Clock Dial

CAD Blueprint: clock_dial

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/clock_quadrants.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White Faceplate with Oceanic Teal Hands (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**How many degrees does minute hand rotate in** Standard: CCSS.MATH.4.MD.C.5	60°	[1][0] [ENTER] [6] [×]
2	**How many degrees does minute hand rotate in** Standard: CCSS.MATH.4.MD.C.5	120°	[2][0] [ENTER] [6] [×]
3	**How many degrees does minute hand rotate in** Standard: CCSS.MATH.4.MD.C.5	300°	[5][0] [ENTER] [6] [×]
4	**A minute hand turns 180°. How many minutes p** Standard: CCSS.MATH.4.MD.C.5	30 minutes	[1][8][0] [ENTER] [6] [÷]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for confusing 60-minute time intervals with decimal tenths (e.g. 1.5 hours is 1 hr 30 min, not 1 hr 50 min).

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts): Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.

Proficient (14–17 pts): Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.

Developing (10–13 pts): Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.

Intervention (<10 pts): Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.